FINITE STANDARD-OPEN SUBCOVERS ADAPTED TO AN AFFINE COVER

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2232-2238, following the statement at 2221-2230.

Editorial note. Quasi-compactness of the affine V is the standard ingredient turning the pointwise construction into a finite cover.

Allowed standard prerequisites (editorial).

- Common standard-open neighborhoods for two affine opens. lemma-standard-open-two-affines (card L26410, cited at ~2234). Produces W_v standard in both V and U_i.
- Standard covering definition. definition-standard-covering (same chapter, cited at ~2228). Fixes the covering class.

Source reference. Original definition 01HT: sources/stacks-schemes/pinned-original-source.tex, source lines 797–797. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01IW: sources/stacks-schemes/pinned-original-source.tex, source lines 2199–2199. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $X = \bigcup_i U_i$ be an affine open covering. Let $V \subset X$ be an affine open. There exists a standard open covering $V = \bigcup_{j=1,\ldots,m} V_j$ (see Definition 01HT) such that each V_j is a standard open in one of the U_i.*

Proof. Pick $v \in V$. Then $v \in U_i$ for some i. By Lemma 01IW above there exists an open $v \in W_v \subset V \cap U_i$ such that W_v is a standard open in both V and U_i. Since V is quasi-compact the lemma follows. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.